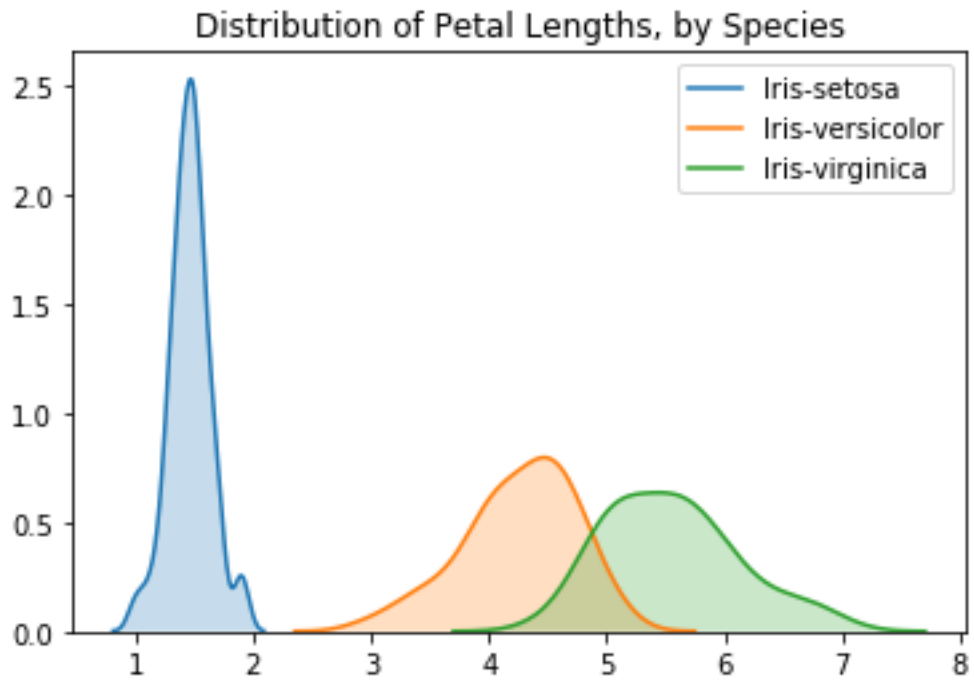
Color-coded plots

```
iris_set_data = pd.read_csv("C:\\datasets_116573_334386_iris_setosa.csv")
iris_ver_data = pd.read_csv("C:\\datasets_116573_334386_iris_versicolor.csv")
iris_vir_data = pd.read_csv("C:\\datasets_116573_334386_iris_virginica.csv")

# KDE plots for each species
sns.kdeplot(data=iris_set_data['Petal Length (cm)'], label="Iris-setosa", shade=True)
sns.kdeplot(data=iris_ver_data['Petal Length (cm)'], label="Iris-versicolor", shade=True)
sns.kdeplot(data=iris_vir_data['Petal Length (cm)'], label="Iris-virginica", shade=True)

# Add title
plt.title("Distribution of Petal Lengths, by Species")
```

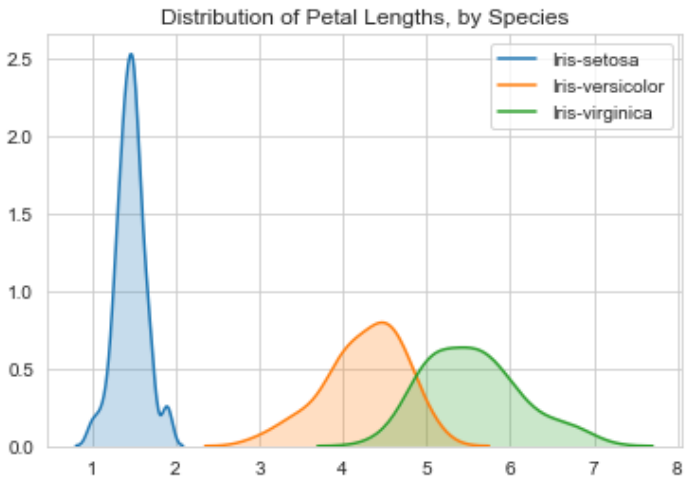

Distribution of Petal Lengths, by Species
Iris-setosa
Iris-versicolor
Iris-virginica
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7
8

Choosing Plot Types

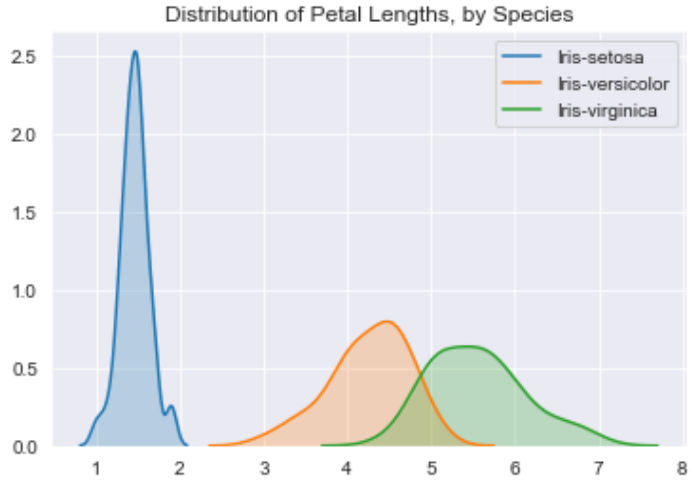
- **Trends** - A trend is defined as a pattern of change.
  - `sns.lineplot` - **Line charts** are best to show trends over a period of time, and multiple lines can be used to show trends in more than one group.
- **Relationship** - There are many different chart types that you can use to understand relationships between variables in your data.
  - `sns.barplot` - **Bar charts** are useful for comparing quantities corresponding to different groups.
  - `sns.heatmap` - **Heatmaps** can be used to find color-coded patterns in tables of numbers.
  - `sns.scatterplot` - **Scatter plots** show the relationship between two continuous variables; if color-coded, we can also show the relationship with a third categorical variable.
  - `sns.regplot` - Including a **regression line** in the scatter plot makes it easier to see any linear relationship between two variables.
  - `sns.lmplot` - This command is useful for drawing multiple regression lines, if the scatter plot contains multiple, color-coded groups.
- **Distribution** - We visualize distributions to show the possible values that we can expect to see in a variable, along with how likely they are.
  - `sns.kdeplot` - **KDE plots** show an estimated, smooth distribution of a single numerical variable (or two numerical variables).

Changing styles with seaborn

Seaborn has five different themes: (1)`"darkgrid"`, (2)`"whitegrid"`, (3)`"dark"`, (4)`"white"`, and (5)`"ticks"`,



sns.set_style("whitegrid")



sns.set_style("darkgrid")